Sub Code:KEC302

Paper Id: 130322 Roll No:

# B. TECH.
## (SEM-III) THEORY EXAMINATION 2019-20
## DIGITAL SYSTEM DESIGN

*Time: 3 Hours* *Total Marks: 100*

**Note:** Attempt all Sections. If require any missing data; then choose suitably.

## SECTION A

**1. Attempt *all* questions in brief.** **2 x 10 = 20**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | The solution to the quadratic equation $k^2$-*11k* + *22* = *0*are x = 3 and x = 6. What is the base of the number system? | 2 | 1 |
| b. | Simplify the expression F (A, B, C, D) = ACD+$\bar{A}$ B + $\bar{D}$ by K- Map. | 2 | 1 |
| c. | Construct half subtractor using logic gates. | 2 | 2 |
| d. | Implement a 4:1 multiplexer using 2:1 multiplexer. | 2 | 2 |
| e. | What do you mean by race around condition in JK Flip Flop? | 2 | 3 |
| f. | Distinguish between Leach and Flip Flop. | 2 | 3 |
| g. | What is logic family? Give the classification of logic families in brief. | 2 | 4 |
| h. | Describe figure of merit & noise immunity of TTL & CMOS ICs. | 2 | 4 |
| i. | What are the advantages and disadvantages of flash type ADC? | 2 | 5 |
| j. | The basic step of a 9-bit DAC is 10.3 mV. If 000000000 represents 0Volts, what is the output for an input of 101101111? | 2 | 5 |

## SECTION B

**2. Attempt any *three* of the following:** **3 x 10 = 30**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Design an excess-3 to BCD code converter. | 10 | 1 |
| b. | Implement a full adder by using 8:1 multiplexer. | 10 | 2 |
| c. | Design a sequential circuit with two Flip Flops, A & B and one input x. When x=0, the State of the circuit remains the same when x=1 the circuit passes through the state transitions from 00 to 01 to 11 to 10 back to 00 & repeat. | 10 | 3 |
| d. | Compare TTL and CMOS logic families and also draw CMOS NOR gate. | 10 | 4 |
| e. | Explain the operation of successive approximation ADC. Discuss it merits and demerits. | 10 | 5 |

## SECTION C

**3. Attempt any *one* part of the following:** **1 x 10 = 10**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Minimize the logic function using Quine-McCluskey Method $F(A, B, C, D, E) = \sum m(8,9,10,11,13,15,16,18,21,24,25,26,27,30,31)$ | 10 | 1 |
| b. | Simplify the logic expression using K-Map $F(A,B,C,D,E,F) = \sum m(0,5,7,8,9,12,13,23,24,25,28,29,37,40,42,44,46,55,56,57,60,61)$ | 10 | 1 |



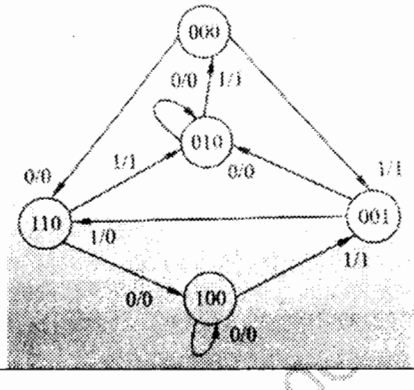
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**4. Attempt any *one* part of the following: 1 x 10 = 10**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Design a 4-bit parallel binary Adder/Subtractor circuit. | 10 | 2 |
| b. | Design a 4-bit comparator circuit using logic gates. | 10 | 2 |

**5. Attempt any *one* part of the following: 1 x 10 = 10**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Discuss Mealy and Moore FSM. What do you mean by excitation table? | 10 | 3 |
| b. | For the given state diagram design the circuit using T flip flop   | 10 | 3 |

**6. Attempt any *one* part of the following: 1 x 10 = 10**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | Draw three input standard TTL NAND gate circuit and explain its operation. | 10 | 4 |
| b. | Implement the following function using PLA<br>$F_1 = \sum m(0,3,4,7)$<br>$F_2 = \sum m(1,2,5,7)$ | 10 | 4 |

**7. Attempt any *one* part of the following: 1 x 10 = 10**

| Qno. | Question | Marks | CO |
|---|---|---|---|
| a. | With a neat diagram explain the operation of R-2R DAC. | 10 | 5 |
| b. | With a neat sketch explain the operation of Flash ADC. | 10 | 5 |

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